

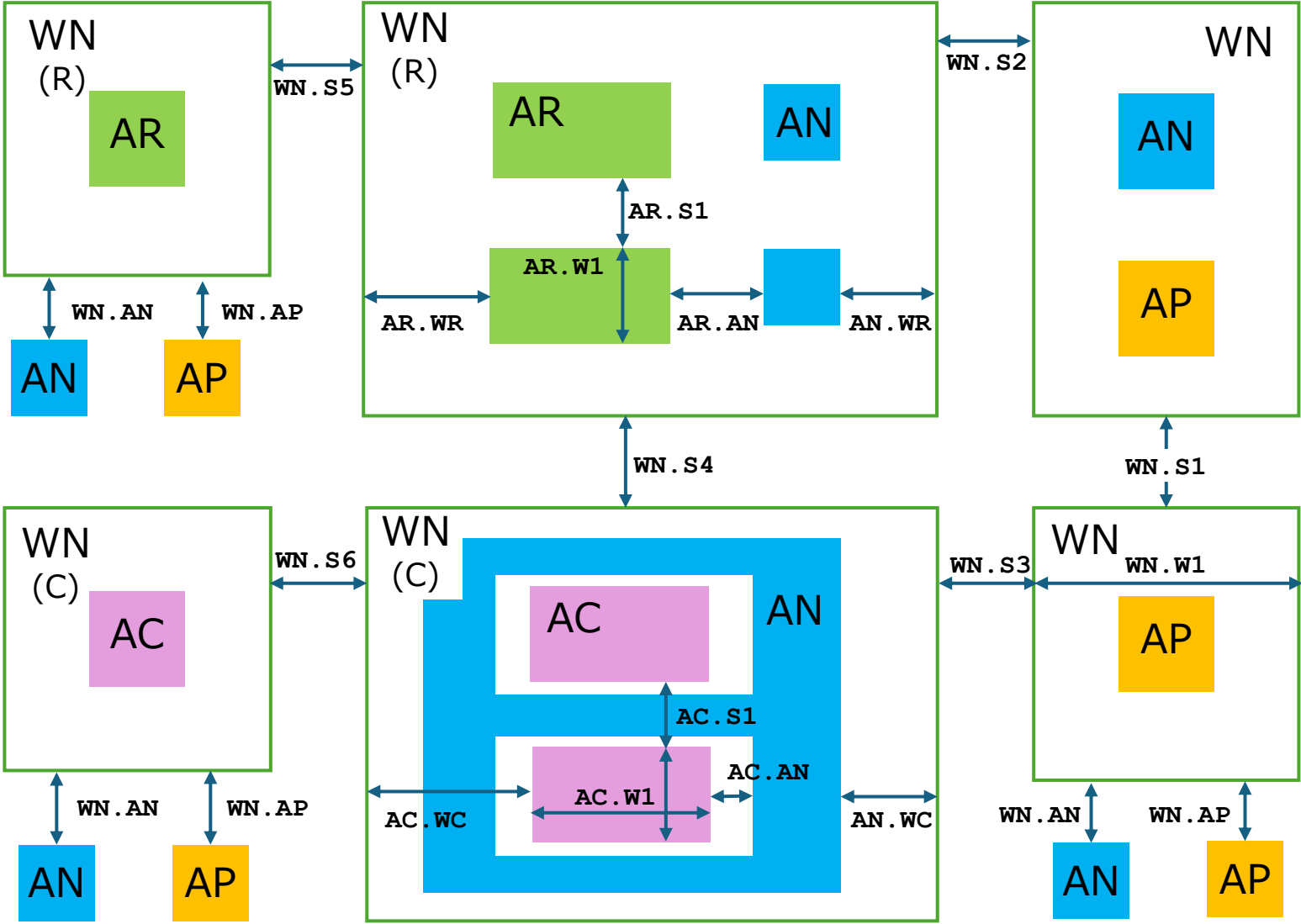
Design Rules

for Drawing Layers

Drawing Layer Table

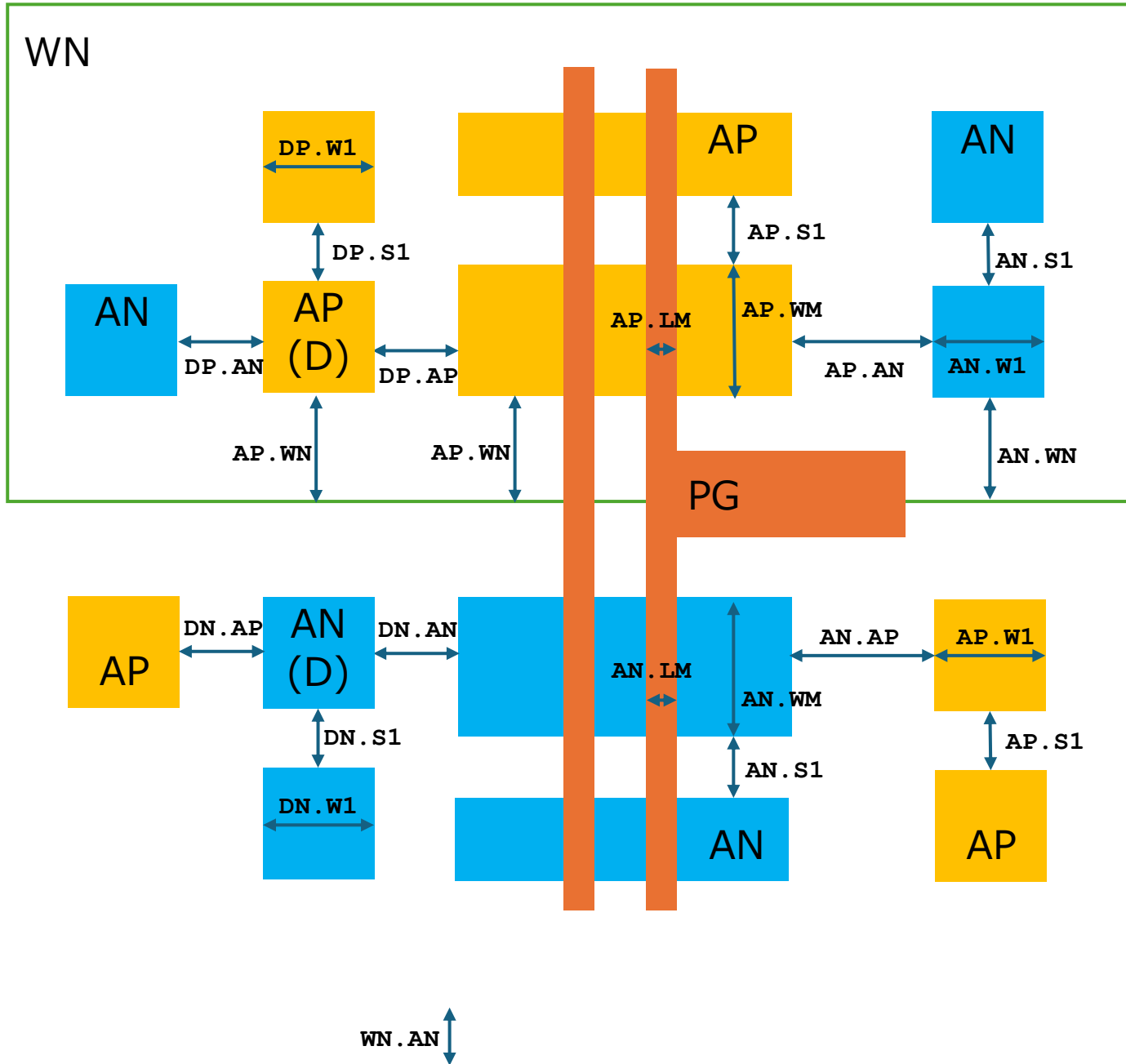
	Original MASK layers				Proposed Drawing layers					
	Name	NUM	TYPE		Name	NUM	TYPE			
1	PSUB	140	0		WN	140	0			
2	NW	36	0					MDP generatioed		
3	HVNW	141	0					MDP generatioed		
4	L	3	0					MDP generatioed		
					AP	3	1	Active for P+		
					AN	3	2	Active for N+		
					AR	3	3	Active for resistor		
					AC	3	4	Active for capacitance		
5	NF	25	0					MDP generatioed		
6	PF	26	0					MDP generatioed		
7	CL	143	0					MDP generatioed		
8	HPBE	144	0					No USE		
9	HNBE	145	0					No USE		
10	PBE	146	0					MDP generatioed		
11	NBE	147	0					MDP generatioed		
12	HPM	33	0					MDP generatioed		
13	RHP	148	0					No USE		
14	SG	8	0							
					PO	8	1	POLY transistor		
					PR	8	2	POLY resistor		
15	PM	35	0					MDP generatioed		
16	NM	7	0					MDP generatioed		
17	R	12	0					MDP generatioed		
18	PSD	9	0					MDP generatioed		
19	NSD	28	0					MDP generatioed		
20	CONT	11	0		CO	11	0			
21	M1	13	0		M1	13	0			
22	TC	19	0		V1	19	0			
23	M2	20	0		M2	20	0			
24	PRO	14	0		PE	14	0			
	TXM1	48	0		TXM1	48	0	M1 Label		
	PIN	48	1		PIN	48	1	M1 Pin		
	TXM2	49	0		TXM2	49	0	M2 Label		
	MASK	63	0		MASK	63	0	Waiver region		
	MEAS	63	1		MEAS	63	1	Measurement region		
	SCRIBE	80	0		SCRIBE	80	0	Waiver region (EDGE SEAL)		
	prBoundary	235	0		prBoundary	235	0	Cell boundary		

WN to AP/AN/AR/AC related



Name	Description	Func	Rule	MAX
WN.W1	WN	Wmin	8.0	inf
WN.S1	WN-WN	Smin	12.0	inf
WN.S2	WN-WN (R)	Smin	9.5	inf
WN.S3	WN-WN (C)	Smin	12.0	inf
WN.S4	WN (R) -WN (C)	Smin	9.5	inf
WN.S5	WN (R) -WN (R)	Smin	8.0	inf
WN.S6	WN (C) -WN (C)	Smin	12.0	inf
WN.AP	WN-AP	Smin	5.0	inf
WN.AN	WN-AN	Smin	10.0	inf
AR.W1	AR	Wmin/max	2.8	20.0
AR.S1	AR	Smin	4.0	inf
AR.WR	AR-WN (R)	Emin	10.0	inf
AN.WR	AN-WN (R)	Emin	5.0	inf
AR.AN	AR-AN	Smin	4.0	inf
AC.W1	AC	Wmin/max	28.5	120.0
AC.S1	AC	Smin	???	???
AC.WC	AC-WN (C)	Emin	???	???
AN.WC	AN-WN (C)	Emin	3.0	inf
AC.AN	AC-AN	Smin	2.8	inf
AC.R1	AN surround AC			

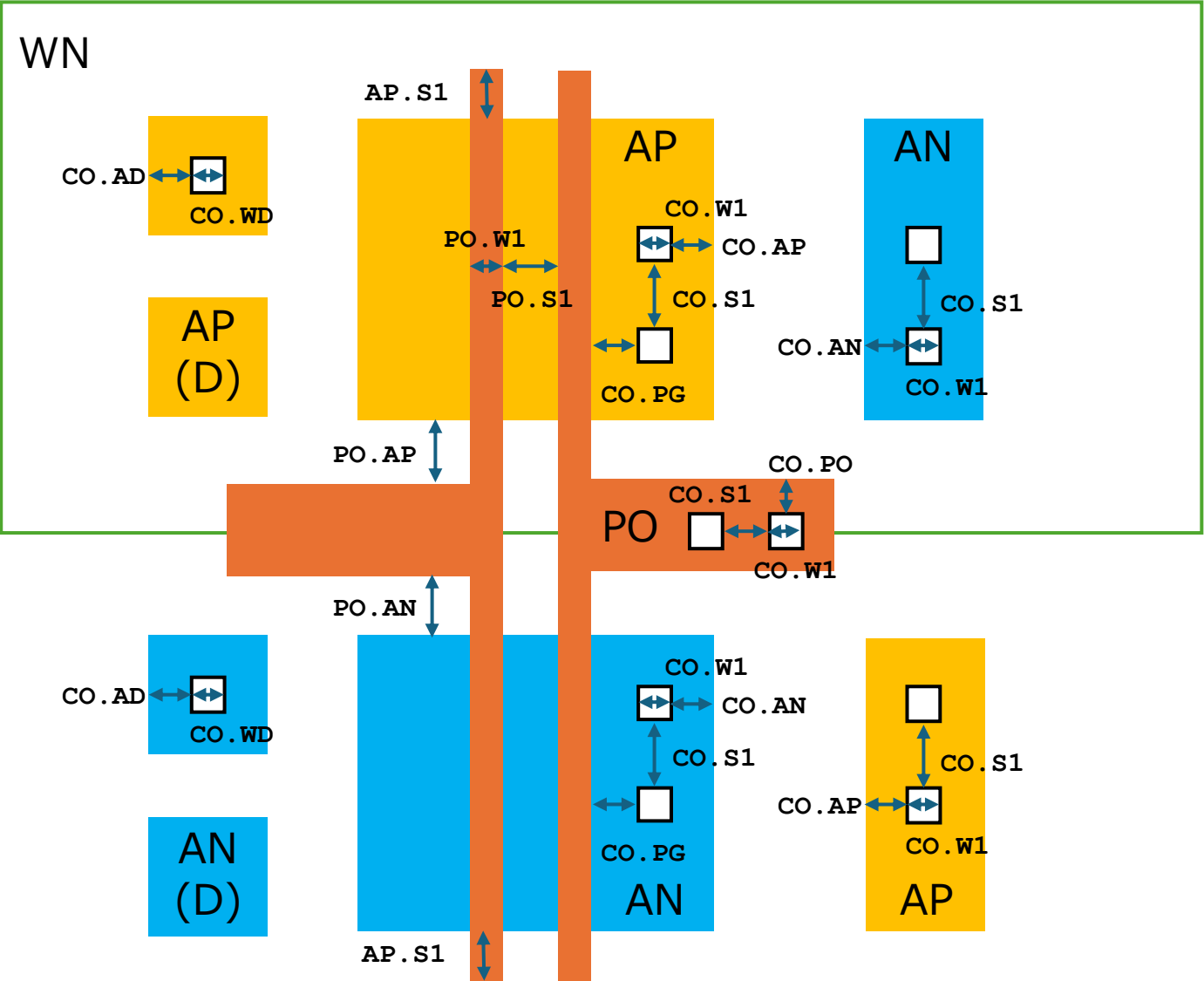
MOS transistor related



Name	Description	Func	Rule	MAX
AP.W1	AP	Wmin	1.4	inf
AP.S1	AP	Smin	1.4	inf
AP.AN	AP-AN	Smin	2.8	inf
AP.WM	PMOS	Wmin/max	3.4	60.0
AP.LM	PMOS	Lmin/max	1.0	30.0
AP.WN	AP-WN	Emin	7.0	inf
DP.W1	AP	Wmin	1.4	inf
DP.S1	DP	Smin	???	???
DP.AP	DP-AP	Smin	2.8	inf
DP.AN	DP-AN	SMin	2.8	inf
AN.W1	AN	Wmin	1.4	inf
AN.S1	AN	Smin	1.4	inf
AN.AP	AN-AP	Smin	2.8	inf
AN.WM	NMOS	Wmin/max	3.4	60.0
AN.LM	NMOS	Lmin/max	1.0	30.0
AN.WN	AN-WN	Emin	5.0	inf
DN.S1	DN	Smin	???	???
DN.AP	DP-AP	Smin	2.8	inf
DN.AN	DP-AN	Smin	2.8	inf

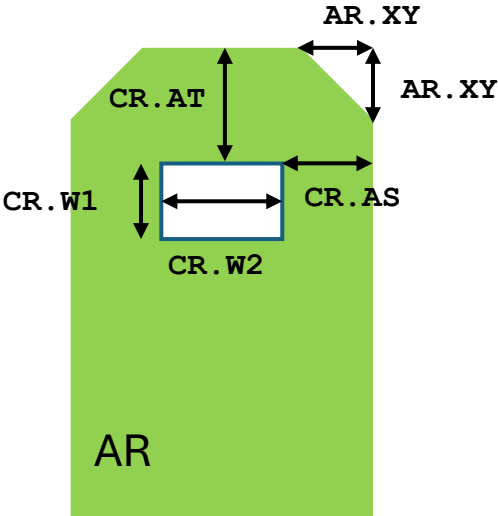
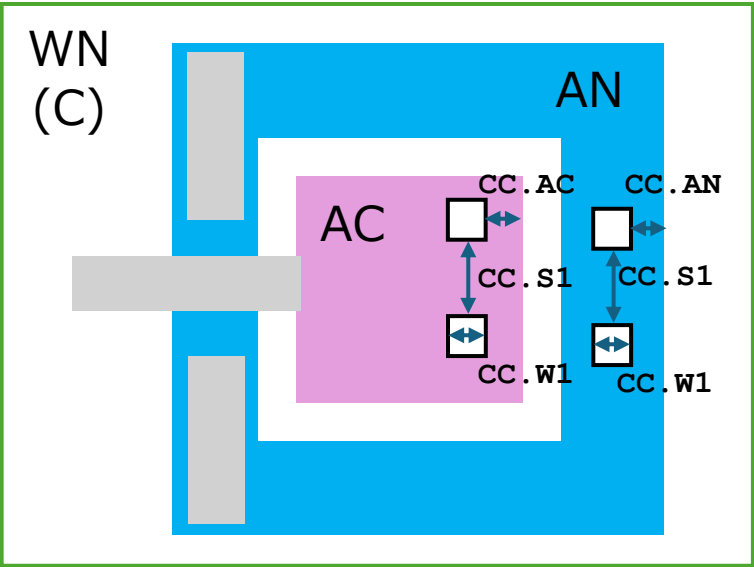
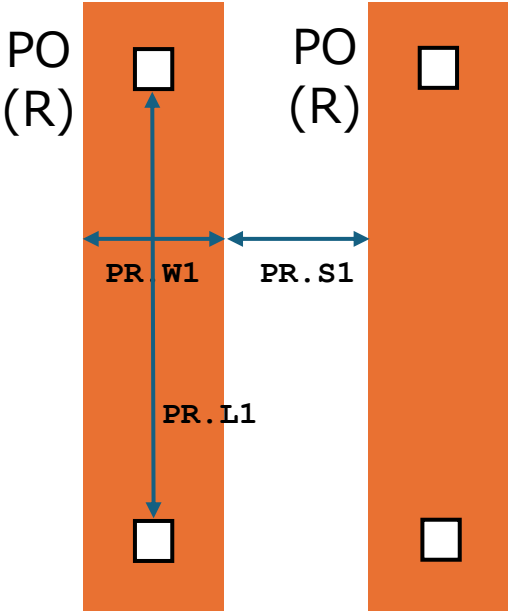
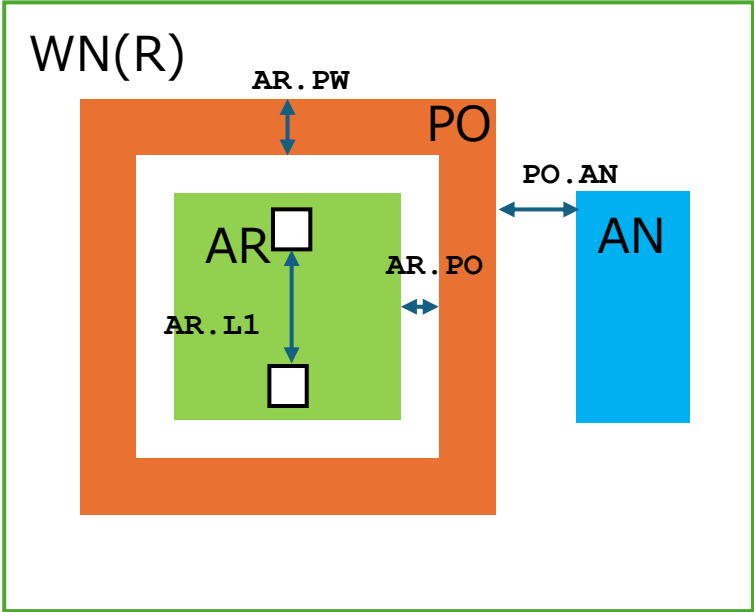
AN . WN

Contact and Poly related



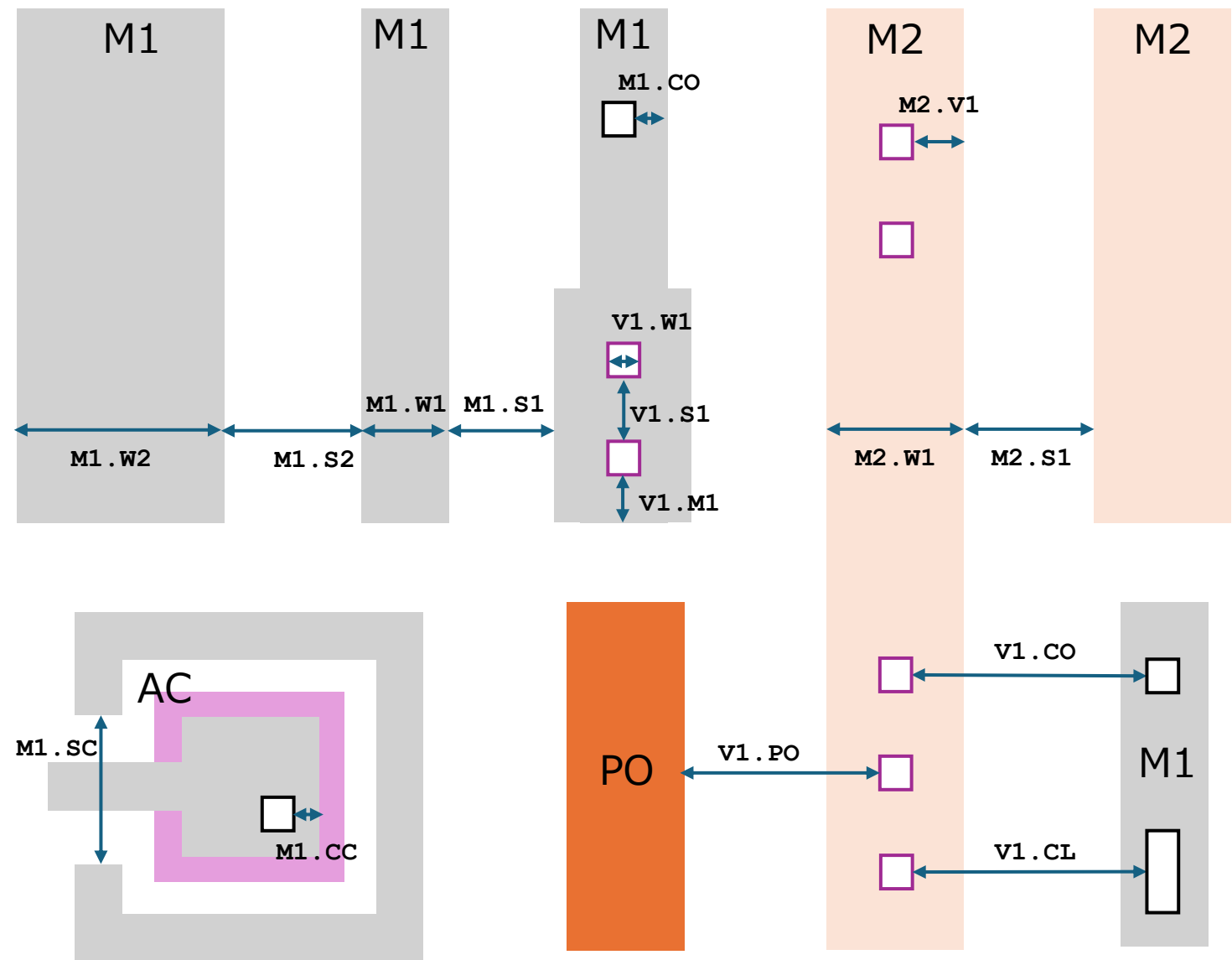
Name	Description	Func	Rule	MAX
PO.W1	PO	Wmin	1.0	30.0
PO.S1	PO	Smin	1.2	inf
PO.AP	PO-AP	Smin	0.4	inf
PO.AN	PO-AN	Smin	0.4	inf
PO.EM	MOS Endcap min	min	1.2	inf
CO.W1	CO	Wmin	1.0	inf
CO.S1	CO	Smin	1.0	inf
CO.WD	CO (D)	Width	1.2	inf
CO.AP	CO-AP	Emin	0.8	inf
CO.AN	CO-AN	Emin	0.8	inf
CO.PG	CO-PO (G)	Smin	1.0	inf
CO.PO	CO-PO	Emin	0.8	inf
CO.AD	CO-AD	Emin	1.2	inf

Resistor(AR/AS) and Capacitor(CSIO) related



Name	Description	Func	Rule	MAX
AR.L1	AR	Lmin/max	13.0	100.0
AR.PO	PO-AR	Smin	1.0	inf
AR.PW	PO (AR)	Wmin	2.0	inf
AR.XY	AR corner cut	XY	1.0	inf
PO.L1	PO (AR)	Lmin/max	20.0	200.0
PO.AN	PO (AR) -AN	Smin	???	???
PO.R1	PO surround AR			
PO.R2	PO tie-down			
PR.W1	PO (RS)	Wmin/max	4.0	20.0
PR.L1	PO (RS)	Lmin/max	20.0	100.0
PR.S1	PO (RS)	Smin	2.0	inf
CR.W1	CO (RR)	Width	1.0	inf
CR.W2	CO (RR)	Wmin	1.0	inf
CR.AT	CO (RR) -AR (Top)	Emin	1.5	inf
CR.AS	CO (RR) -AR (Side)	Emin	1.0	inf
CC.W1	CO (C)	Wmin	1.2	inf
CC.S1	CO (C)	Smin	1.2	inf
CC.AC	CO (C) -AC	Emin	2.9	inf
CC.AN	CO (C) -AN	Emin	1.2	inf

M1 /V1 /M2 related



Name	Description	Func	Rule	MAX
M1.W1	M1	Wmin	1.8	inf
M1.S1	M1	Smin	1.4	inf
M1.SC	M1 (C)	Smin	10.0	inf
M1.CO	CO-M1	Fmin	0.8	inf
M1.CC	CO (C) -M1	Fmin	1.2	inf
M1.WW	M1 (W)	Wmin	10.0	inf
M1.SW	M1 (W)	Smin	2.0	inf
V1.W1	V1	Width	1.4	inf
V1.S1	V1	Smin	1.5	inf
V1.M1	V1-M1	Emin	1.0	inf
V1.PO	V1-PO	Smin	1.2	inf
V1.CO	V1-CO	Smin	1.0	inf
V1.CL	V1-CO (L)	Smin	1.4	inf
M2.W1	M2	Wmin	3.0	inf
M2.S1	M2	Smin	2.0	inf
M2.V1	V1-M2	Fmin	1.0	inf

質問・確認事項

2025/12/25

DIODE

リファレンス・マニュアルTable I-1-1によればダイオードはP/NどちらもPM+NMインプラと記載

DP : PM+NM

DN : PM+NM

リファレンス・マニュアル図6.1/6.2にも同様の指示があるが、

リファレンス・マニュアル図6.12/6.13に(ESD)では

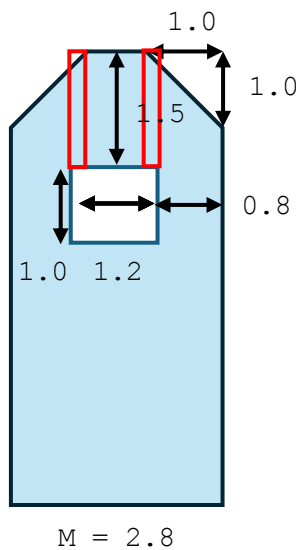
DP : NM

DN : PM

と指示がある。本当はどちらでしょう？違う場合は、Spiceモデルを公開頂けると幸いです。

また、ESD用NMOSのSPICEモデルも公開頂けると幸いです。

Design Rule for RR details



1. RR抵抗幅2.8umの場合：

ERRR14: 0.8um

RRのときは可変長コンタクト

ERRR08 : 1.5um

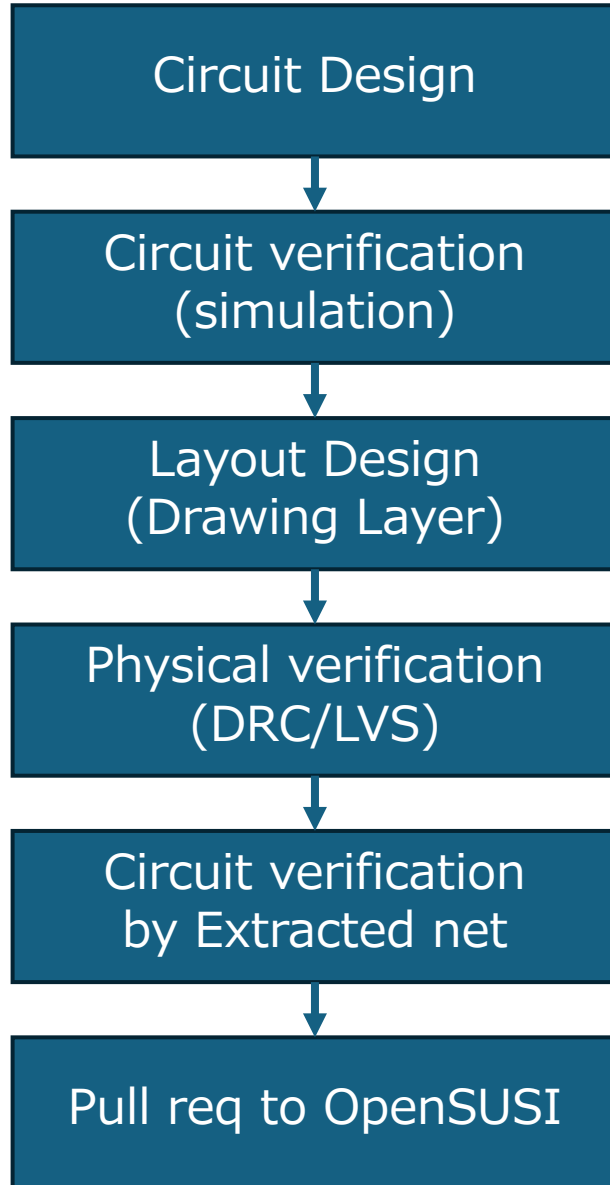
赤枠のエリアで ERR08 がエラー

* 角カットを1.0>0.8に変更しては？

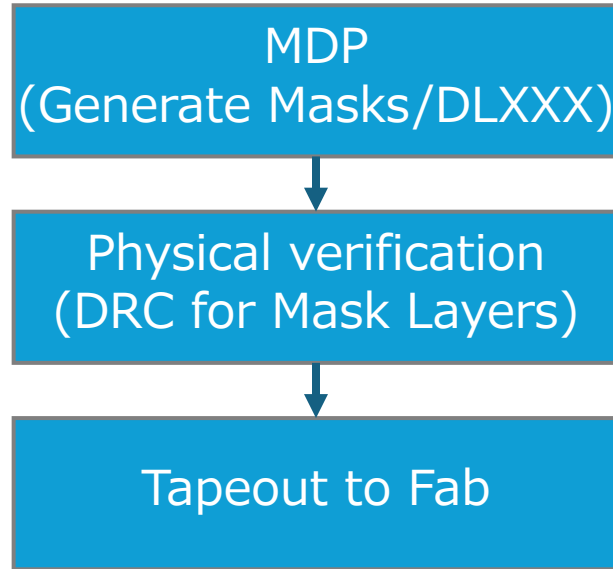
Design to Tapeout Process

Mask Data Preparation

Designer



OpenSUSI



Fab

